



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

May 05, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 11650
(THIRTY-FIRST REVISION)

EXPIRATION DATE: 2029-04-30

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Autoliv ASP, Inc.
Ogden, UT

2. PURPOSE AND LIMITATIONS:

a. This special permit authorizes the manufacture, mark, sale, and use of non-DOT specification pressure vessels for use as components of airbag inflators or seat-belt pretensioners when they are articles of commerce in transportation. The pressure vessels, charged with non-toxic, non-liquefied gases, are authorized for transportation in commerce subject to requirements and limitations specified herein. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.

b. FIVE-YEAR TRANSPORTATION AUTHORIZATION: This special permit authorizes transportation of the pressure vessels identified herein for up to five years from the date of manufacture. This limitation does not apply to non-specification pressure vessels when installed in air bag modules. This special permit provides no certification of safety for end use environments and life cycles.

c. SPECIAL PERMIT SCOPE LIMITATIONS: This special permit only applies to a package of pressure vessels, airbag inflators, or seat-belt pretensioners when it is an article of commerce in transportation. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, incorporation as a component of a vehicle or other device, or other uses not associated with transportation in commerce.

d. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.

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3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.301(a)(1) and 173.302(a) in that non-DOT specification cylinders are not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Autoliv ASP, Inc.'s application dated January 21, 2025, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Safety devices	9	UN3268	N/A

Note: A safety system or component that contains a quantity of pyrotechnic materials must be classed and approved as provided for in § 173.56 of the Hazardous Materials Regulations (HMR). If the pyrotechnic material augments the volume of the gas in the pressure vessel, or in any way enhances the performance of the compressed gas, the device must be tested in the same configuration as when shipped.

7. SAFETY CONTROL MEASURES:
- a. PACKAGING: Prescribed packaging is a non-DOT specification pressure vessel designed and manufactured in accordance with the DOT Specification 39 (§§ 178.35 and 178.65), except as follows:
- (1) The maximum service pressure at 70 °F may not exceed 10,200 psig. The rated service pressure may not exceed 80 percent of the test pressure and the water volume of each pressure vessel may not exceed 0.8 liter.
- (2) Material of construction must conform to all requirements of § 178.65(b), except that for inflators up to 5,000 psig service pressure, a maximum ladle check analysis carbon content of 0.20 percent is authorized. The chemical composition and mechanical strength properties must conform to documents on file with Office of Hazardous Materials Safety (OHMS).
- (3) Manufacturing requirements must conform to all requirements of § 178.65(c).

(4) The minimum wall thickness must be such that the wall stress meets the requirements of § 178.65(d).

(5) Openings and attachments must conform to all requirements of § 178.65(e) except that:

(i) Sidewall attachments along the largest outer diameter of the cylinder are authorized provided the attachments are not detrimental to the cylinder's structural integrity. The burst performance of the cylinder shall not be less than 95% of the same cylinder design without attachment. In all cases, the burst pressure must meet the criteria of paragraph 7.b.(1). This must be demonstrated through certification by the Independent Inspection Agency as provided in paragraph 8.d.;

(ii) A sidewall opening, as shown in Autoliv ASP, Inc. drawing 850008 dated February 28, 1996, drawing E009267 dated December 5, 2000, drawing E024437 dated May 28, 2002, or drawing E187383, dated September 21, 2007, on file with OHMS, is authorized;

(6) Each pressure vessel must be equipped with a pressure relief device designed to meet all the requirements for a rupture disk prescribed in the Compressed Gas Association (CGA) Pamphlet S-1.1. The pressure relief device must be capable of preventing rupture of the pressure vessel when subjected to fire test conducted in accordance with CGA Pamphlet C-14.

(7) The pressure vessels may only be charged with non-toxic, non-liquefied gases.

(8) Pressure vessels must be transported in strong outside packaging in accordance with § 173.301(a)(9), unless otherwise directed by § 173.166 or § 173.62.

b. TESTING:

(1) Each pressure vessel must be tested as required in § 178.65(f) except that:

(i) The hold time at test pressure specified in § 178.65(f)(1) may be limited to that which is adequate to ensure compliance with the requirements contained in § 178.65(f)(1); and

(ii) Cylinders manufactured in accordance with ACH-2.4 must be designed to have a minimum burst pressure of 2.7 times the service pressure; and all other cylinders must have a minimum burst pressure of 2.5 times the service pressure; and

(iii) The maximum duration of the shift specified in § 178.65(f)(3) may be extended beyond 10 hours at the discretion of the independent inspector.

(iv) For inflators utilizing ACH-4.1 technology, the lot must be rejected if, during the pressure test, a failure occurs at a gage pressure less than 1.6 times the test pressure. Autoliv ASP, Inc. must submit the test data on the first 5 lots of inflators utilizing ACH-4.1 technology to OHMS prior to the shipment of the lot.

(2) The flattening test specified in § 178.65(g) is not required.

(3) A representative pressure vessel, packaged as it would be for shipment, must be activated and no materials other than non-toxic, non-flammable vapors or gases may be expelled from the package.

(4) A cylinder with an outside diameter less than 2 inches may burst at a location other than that specified in § 178.65(f)(2)(ii) and (iii).

c. MARKING:

Each pressure vessel must be durably marked as follows:

DOT-SP 11650/10,200¹

lot No. xxxxx²

Manufacturer's Name

This Pressure Vessel May Not Be Refilled

¹ Where 10,200 represents the design service pressure.

² Where xxxxx is the lot number as appropriate.

Note: Each line of these markings may be placed without regard to location or order on the pressure vessel.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification

or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

- c. This special permit is limited to pressure vessels used as components for safety systems and explosive articles. The pressure vessels are excepted from the requirements of the HMR, Part 178 when the design has been certified by an Independent Inspection Agency, approved under § 107.803 as having met all the requirements of this special permit.
 - d. The Independent Inspection Agency's design certification must include test results and documents related to explosive classification and approval. A copy of the certification must be maintained at each facility where the pressure vessel is manufactured and by the Independent Inspection Agency for a period of 15 years from the date of completion of the design certification.
 - e. A current copy of this special permit must be maintained at each facility where the pressure vessel is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - f. Autoliv ASP, Inc. must comply with all provisions of this special permit and all other applicable requirements contained in the HMR, Parts 100-180. No modifications may be made to the pressure vessel or pyrotechnic components that would affect the performance of the pressure vessel or its compliance with the requirements of this special permit until such modifications have been reviewed, tested and certified by an Independent Inspector as meeting the requirements of this special permit.
 - g. Devices utilizing the non-DOT specification pressure vessel authorized herein are exempt from the requirements of 49 CFR Parts 100-180 when installed in a motor vehicle or in completed vehicle components such as steering columns or door panels.
- 9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, passenger-carrying aircraft (may not exceed the quantity limitation specified in the § 172.101 Table, Column (9A)), and cargo-only aircraft.
 - 10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft, or motor vehicle used to transport packages covered by this special permit. The shipper shall furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.
 - 11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BrianM/Casey Chambers